

L12b: Why Do Markets Crash?

CHEME 5660 Cornell University

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Objectives for Today

Today, we build a simple agent-based market and ask whether changing one entry of a trading rule can turn a calm market into a falling one.

Today's concepts

- Rules as policies: a lookup table from what neighbors intend to what a trader intends next.
- The dealer: how net buying raises the price, net selling lowers it, and why a switched market needs a control.
- Rule-switch simulation: reading the selling fraction and the price, and what the comparison does not tell us about real crashes.

What You Need to Know

Core concepts

- A Wolfram rule maps the neighbor sum $c \in \{0, 1, \ldots, 8\}$ to the trader's next intention. Rules 19280 and 19286 differ only at $c = 1$: Buy versus Sell.
- The dealer sets the price from the net trade, so we compare every switched market with a control that shares its history.
- Giving the changed rule to a fraction f of the traders can tip the market toward a Sell majority while the control does not. This is an analogy for crashes, not an explanation of them.

For understanding (you won't reproduce these)

- The dealer's price update formula and the technical traders' band rule.
- The edge-trader conditions and the Q-learning notebook.

Lectures and Examples

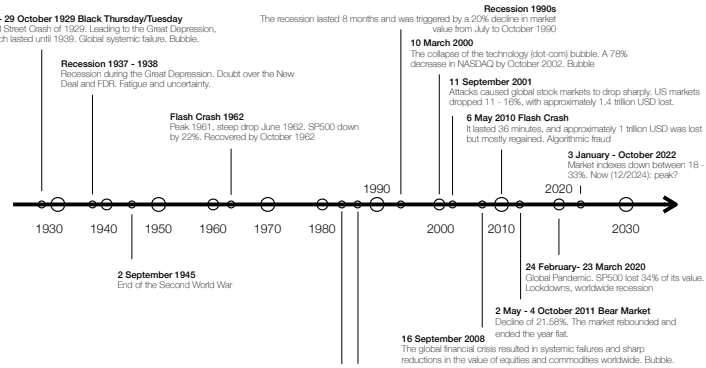
Lecture:
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Example:
CHEME-5660-L12b-Example-Wolfram-RuleSwitch-Fall-2026.ipynb

Optional advanced notebooks are listed on the Optional Advanced Material frame.

Why Do Markets Crash?

A market crash is a sudden, significant decline in prices across a broad market. Three episodes show investor psychology swinging from euphoria to panic:



Three Landmark Episodes

Three episodes, each with a different trigger:

- 1929 Wall Street Crash: margin-fueled speculation met a sudden loss of confidence, and leveraged investors rushed to cover.
- 1987 Black Monday: portfolio insurance and program trading amplified a slide, and a one-day 22.6% Dow drop followed.
- April 2025 tariff mini-crash: tariff news and fast algorithmic selling hit sentiment first and fundamentals second.

Sources: Federal Reserve History; Brady Report; USTR

Agent-Based Modeling of Market Crashes

Markets are man-made systems, and they can fail. An agent-based model simulates traders as autonomous agents with their own rules, and the price emerges from their actions:

- Iori (2002): interacting traders who imitate each other produce large fluctuations and volatility clustering.
- Lux–Marchesi (1999): agents switch between fundamentalist and chartist strategies, producing bubbles and crashes.

Today, traders sit on a grid and act on what their neighbors intend to do.

Sources: Iori; Lux and Marchesi, Nature 397

Agents as Policies

A policy is a table that gives, for each possible state of an agent's surroundings, the action the agent takes. In our market:

- State: a number summarizing what a trader's four neighbors intend.
- Action: the trader's own next intention, Buy, Hold, or Sell.
- Source of the table: Q-learning could learn it from experience. Our traders are handed their tables, and we change one entry.

Wolfram Markets

Traders sit on the cells of a square grid, and simple local rules produce market-level behavior, as in Wolfram's cellular automata:

- Intention: Buy, Hold, or Sell, coded 0, 1, 2. We write $z_j(t) \in \{0, 1, 2\}$ for the intention of trader j at step t.
- Neighbors: an interior trader has four, above, below, left, and right, and updates its intention from theirs.

This is a three-color two-dimensional cellular automaton.

Source: our paper on rule switches in Wolfram markets; Wolfram, *A New Kind of Science* (2002), Chapter 3

The Neighbor Sum

Let ∂i denote the four neighbors of trader i. The trader looks only at the neighbor sum $c_i(t)$, not at its own current intention:

$$\underbrace{c_i(t)}_{\text{neighbor sum}} = \sum_{j \in \partial i} z_j(t) \in \{0, 1, \ldots, 8\}$$

The policy is a table with one action $a_c \in \{0, 1, 2\}$ for each of the nine sums $c = 0, 1, \ldots, 8$: the intention the trader adopts next when its neighbor sum is c.

Rule Numbers

There are $3^9 = 19{,}683$ such tables. We identify each by its rule number, the table written in base 3, with digits from a_8 on the left to a_0 on the right:

$$\underbrace{\sum_{c=0}^{8} a_c \, 3^c}_{\text{rule number}}$$

We use Rule 19280, $(222110002)_3$, as the baseline.

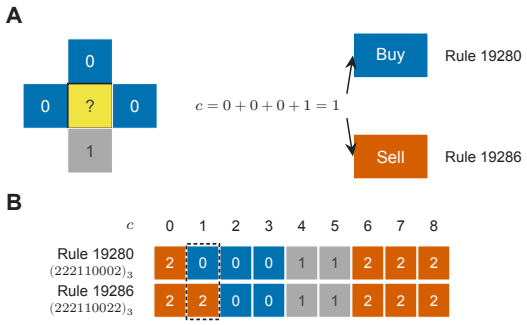
Edge Traders and One Simulation Step

Some edge traders are technical traders: they Buy when the dealer's price falls below a band around its recent average, Sell above it, and Hold otherwise. The others stay in Hold. Each step runs in this order:

1. Rule changes and orders: apply any scheduled rule change; traders order from their intentions.
2. Clearing: the dealer matches orders and updates the price.
3. Technical traders: update their bands and intentions.
4. Interior traders: all update together from the previous intentions of their neighbors.

The Rule Switch

Rule 19286, $(222110022)_3$, differs from the baseline in one entry. Three neighbors intend Buy and one Hold, so $c = 1$: Rule 19280 buys, and Rule 19286 sells. Base-3 digits run from $c = 8$ down to $c = 0$, the table columns run $c = 0$ to 8:



The Rule-Switch Experiment

We use the one-entry difference as a model of a change in psychology. The experiment has three parts:

- Shared history: all interior traders follow Rule 19280. At the switch step, a fraction f of them, chosen in advance, receive Rule 19286 and keep their intentions, cash, and shares.
- Control: a second market with the same history never switches, so differences after the switch come from the rule change.
- Edge traders: in the active condition they react to the price; in the held condition they stay in Hold, so any cascade must come from traders reading their neighbors.

Example: Rule Switch

Example:
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In this example, we plot the fraction of traders who intend to sell and the dealer's price for a switched market beside its control.

The example uses the dealer's price throughout, so next we look at how the dealer sets it.

Prices and the Dealer

The dealer holds cash and shares and takes the other side of trades the traders do not match. At step t, x_t is the net number of shares the dealer sells to traders (positive for net buyers), Q_{t-1} the dealer's shares before the trade, and P_{t-1} the price before it. The price after the trade is:

$$P_t = P_{t-1}\Big(\underbrace{\frac{Q_{t-1}}{Q_{t-1} - x_t}}_{\substack{\text{shares before} \\ \text{over shares after}}}\Big)^2$$

Net buyers, $x_t > 0$: the ratio exceeds one and the price rises. Net sellers, $x_t < 0$: the ratio is below one and the price falls.

Dealer Depth and Public Intentions

- Depth: the dealer's depth h is its initial shares per trader. A deeper dealer holds more shares Q_{t-1}, so a net order moves the price less.
- Public intentions: neighbors read an intention without checking holdings, so it can spread even when its sender cannot trade. A completed trade needs cash for a buyer and a share for a seller.
- Control: a shallow dealer can crash the price with no switch at all, so we read a switched price beside its control's.

Details: the full clearing rule is in the example's code and in the paper

Back to Real Crashes

Our market is a toy, and we do not claim it explains any real crash. The rule switch is a loose analogy:

- 1929: buying neighborhoods ($c = 1, 2, 3$) sustain themselves; the switched entry sells when one neighbor holds back, confidence turning.
- 1987: portfolio insurance told some investors to sell; here a fraction f of traders receive a different entry.
- April 2025: news changed how traders read each other, so the same neighborhood leads to a different action.

Optional Advanced Material

Q-learning: advanced/qlearning/
CHEME-5660-L12b-Advanced-QLearning-Fall-2026.ipynb

How could a trader learn a policy table from experience instead of being handed one? We write the Q-learning update, lay out the algorithm, and describe when it converges to the best policy. Not a prerequisite for L13a.

Summary

We built a simple agent-based market and used it to ask how a small change in trader behavior can tip a market toward selling.

Key takeaways

- Rules as policies: a rule is a lookup table, and one entry can change.
- Dealer and control: net trades set the price, and a control isolates the switch.
- Tipping, by analogy: a switch for some traders can tip the market toward selling, but a simulation does not establish real causes.

Agent-based models ask how individual rules add up to market behavior.